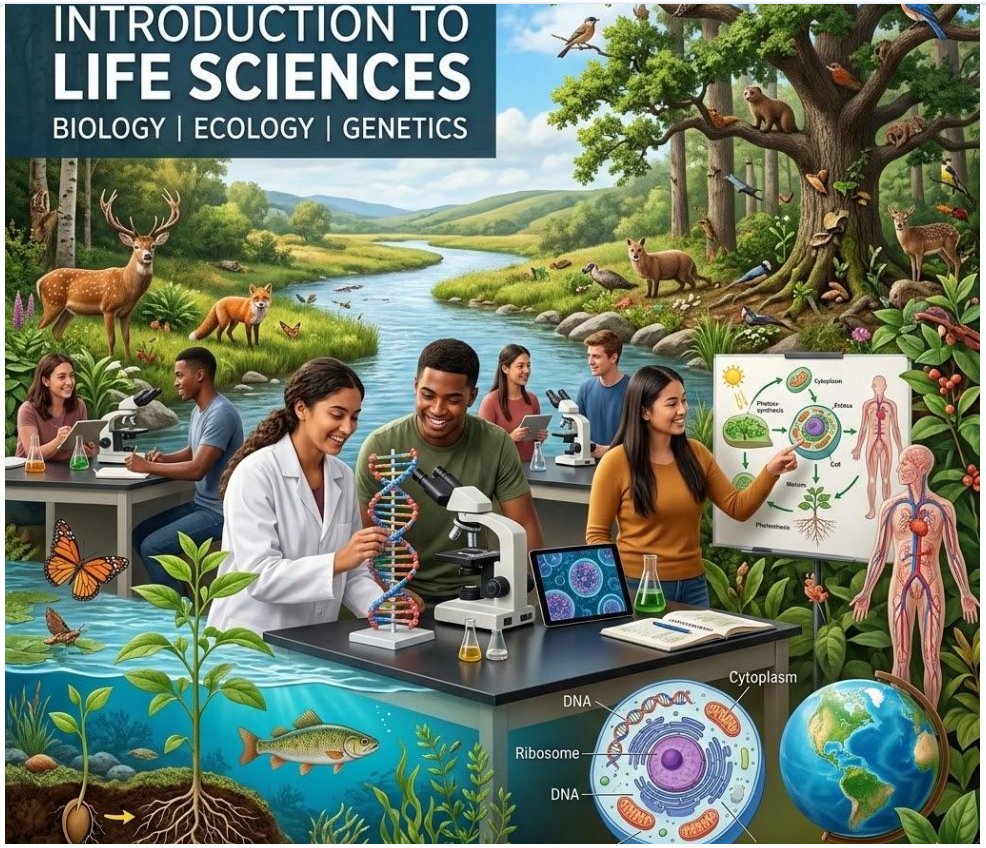
IIT Hyderabad • Department of Biotechnology

BT1010: Introduction to Life Sciences

Lecture 1: Introduction to Life Sciences, Biomolecules: Carbohydrates and Lipids



Course Instructor

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Associate Professor

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Course Schedule – BT1010-Oct-Nov 2026

Lecture/ Faculty	Date/Day	Topic
1 (GM)	12/10/2026 (Mon)	Introduction to Life Sciences_Biomolecules: Carbohydrates and Lipids
2 (SD)	14/10/2026 (Wed)	Human Health and Diseases
3 (SD)	15/10/2026 (Thu)	Human Health and Diseases
4 (SD)	26/10/2026 (Mon)	Immune System
5 (SD)	28/10/2026 (Wed)	Human Nutrition
6 (SD)	29/10/2026 (Thu)	Cancer
7 (GM)	2/11/2026 (Mon)	Biomolecules: Proteins and Enzymes
8 (GM)	4/11/2026 (Wed)	Biomolecules: Nucleic Acids and DNA Based Information Technologies
9 (GM)	5/11/2026 (Thu)	Cell Organelles and Nuclear Organization
10 (GM)	9/11/2026 (Mon)	Techniques used in Cell and Molecular Biology
11 (GM)	11/11/2026 (Wed)	Metabolism and cellular energetics
12 (GM+SD)	12/11/2026 (Thu)	Quiz (25 points) MCQs

Instructor's Biodata

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Education:

2003-2006	B.Sc. Biochemistry, St. Xavier's College, Ahmedabad, Gujarat, India
2007-2009	M.Sc. Microbiology, M.S. University of Baroda, Vadodara, Gujarat, India
2009-2015	Ph.D. Biosciences and Bioengineering, IIT Bombay, Mumbai, Maharashtra, India

Professional Experience:

2015-2020	Post-Doctoral Fellow at the National Institutes of Health (NIH), MD, USA

Research Interests:

Chromosome Dynamics, Gene Regulation, Single-Molecule Imaging, Genetic Disorders, Cancer

Instructions

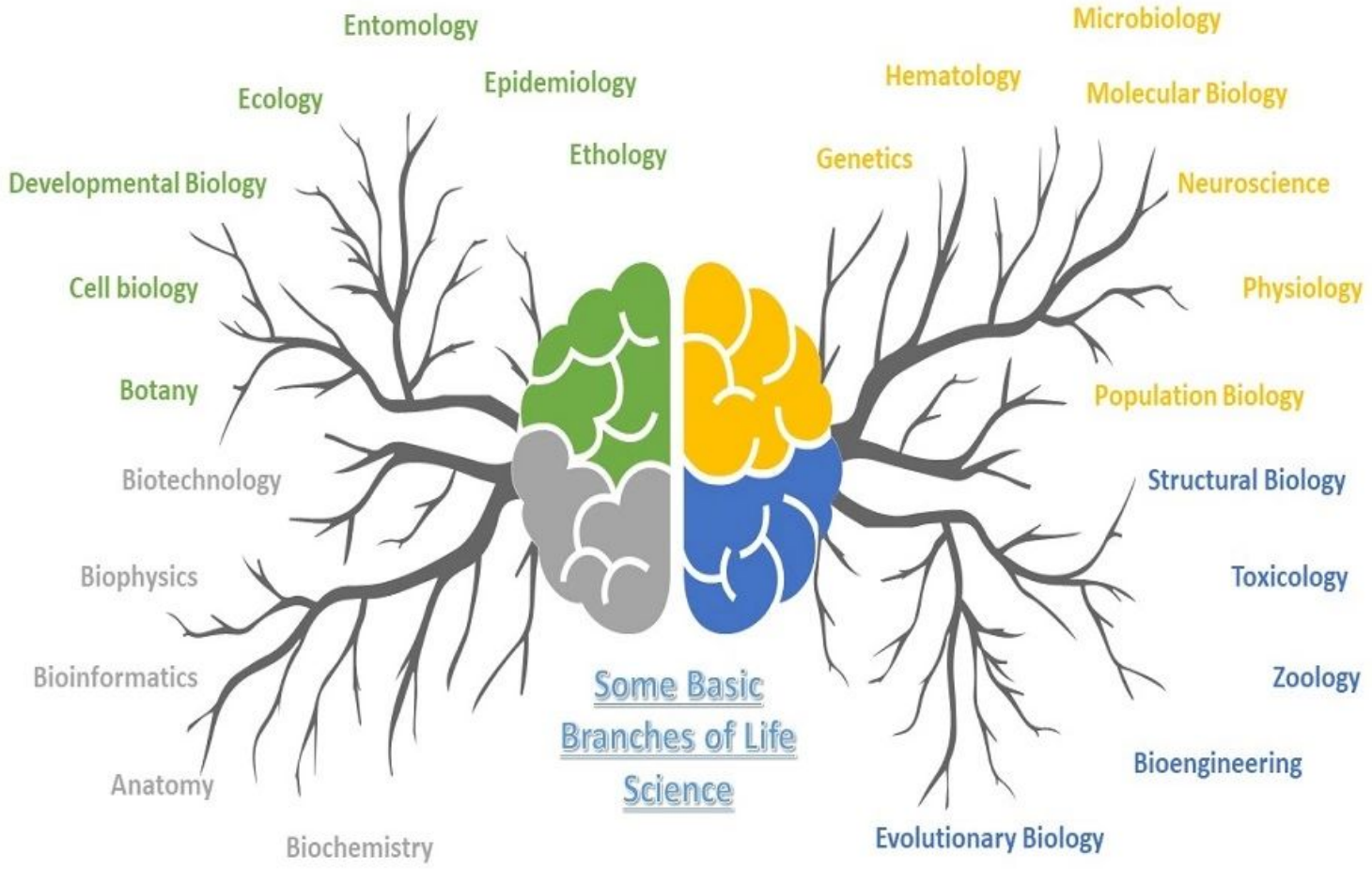
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- Don’t circulate any course material outside (to any non-registered person).

Introduction to Life Sciences (Biology)

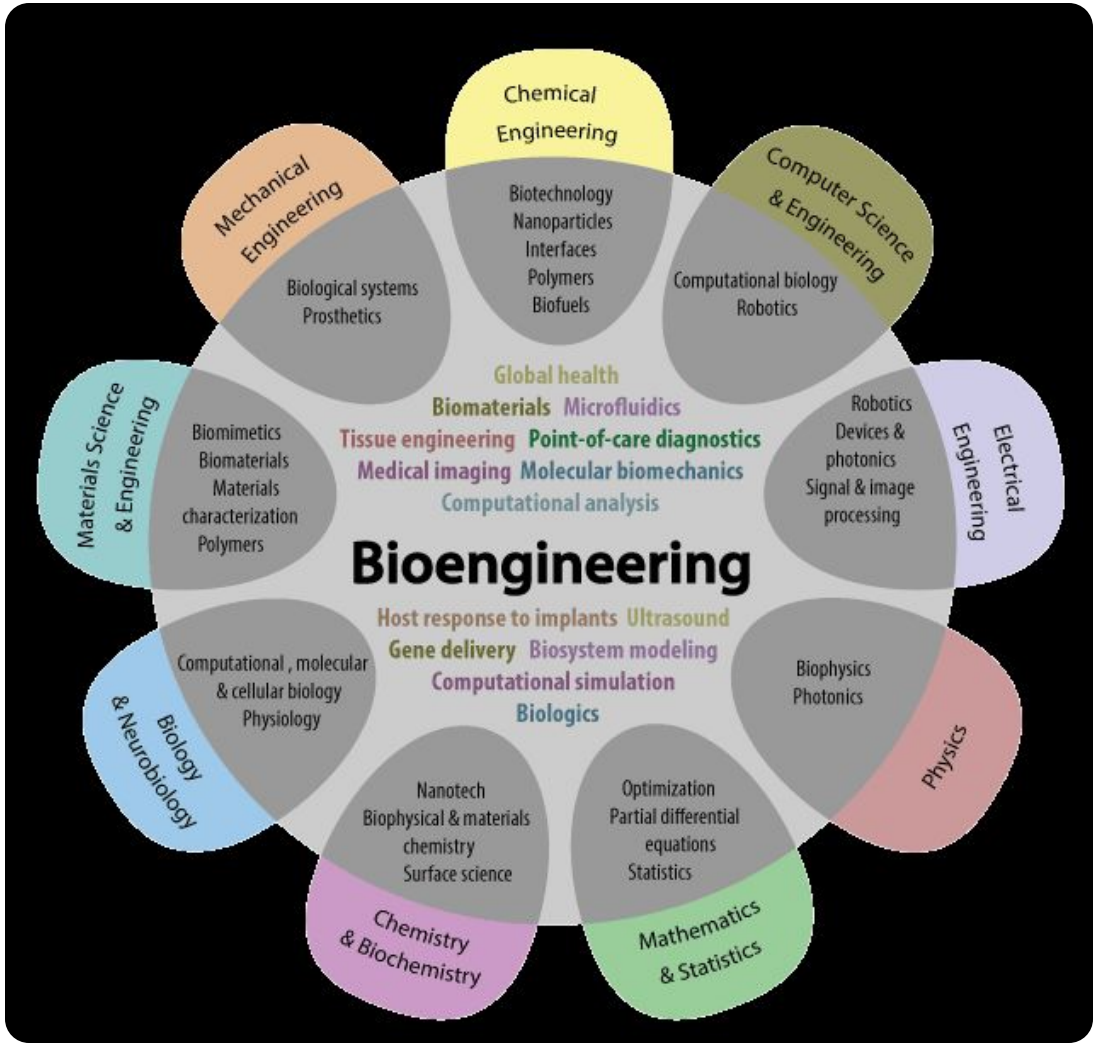
(zoology, botany, biochemistry, microbiology, biotechnology, molecular biology, genetic engineering, genetics etc.)

- Life science studies life in all its forms, past and present.
- This can include plants, animals, viruses and bacteria, single-cell organisms, and even cells.
- Life sciences study the biology of how these organisms live, which is why you may hear this group of specialties referred to as biology.
- The simplest way to define life sciences is the study of living organisms and life processes.

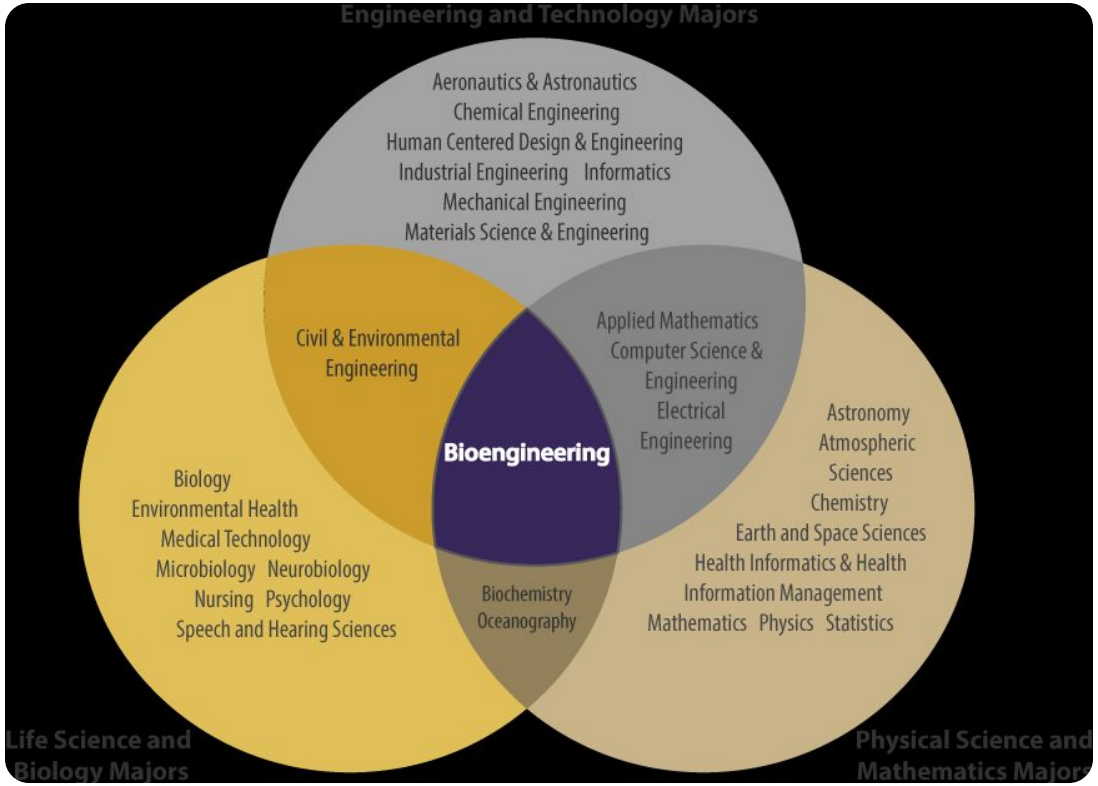


Why Engineers Need to Study Biology / Life Sciences

Interdisciplinary Bioengineering Map



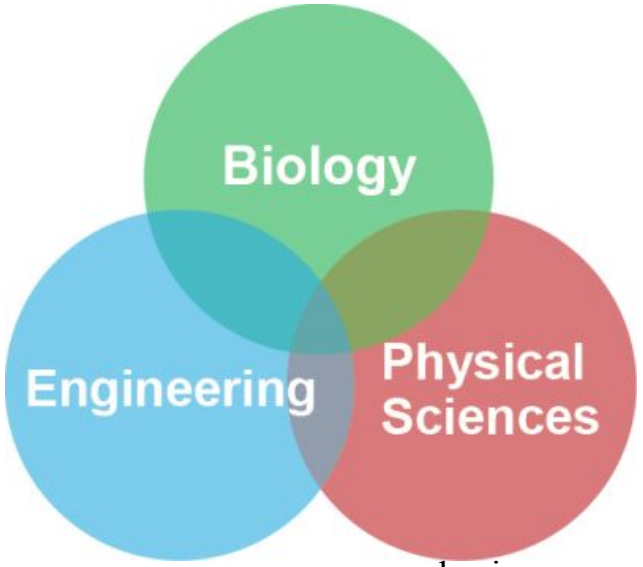
Convergence of Core Disciplines



Source Reference:
https://bioe.uw.edu/academic-programs/about-bioengineering/

How Biology/Biomedical Engineering is linked to other disciplines of Engineering?

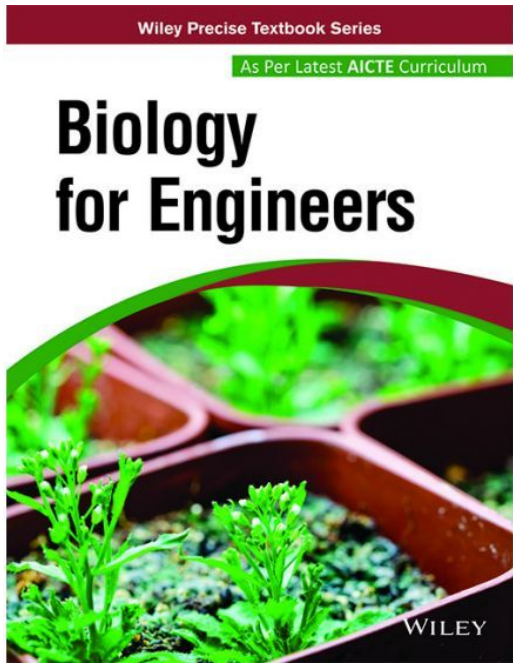
Biomedical Engineering	*Mechanical Engineering*	Biomechanics
		Biofluid
		Bionics
		Rehabilitation Engineering
		Orthopaedic Bioengineering
	Electrical and Computer Engineering	Biomedical Electronics
		Biomechatronics
		Bioinstrumentation
		Medical Imaging
		Systems Physiology
		Neural Engineering
		Bioinformatics
		Clinical Engineering
	Biological, Chemical, Material Sciences	Biomaterials
		Cellular and Tissue Engineering
		Genetic Engineering

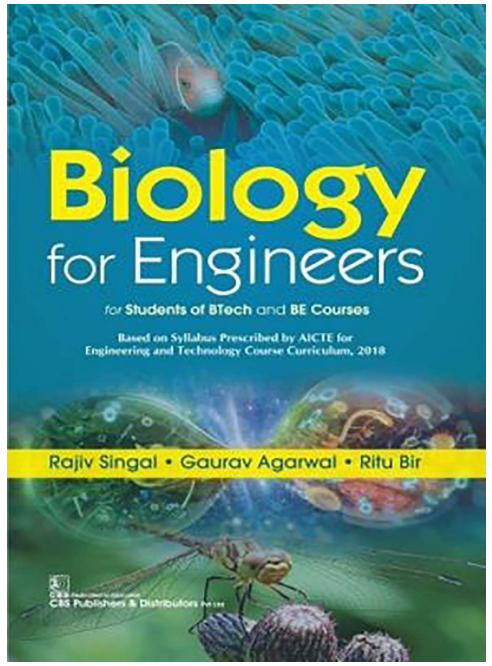


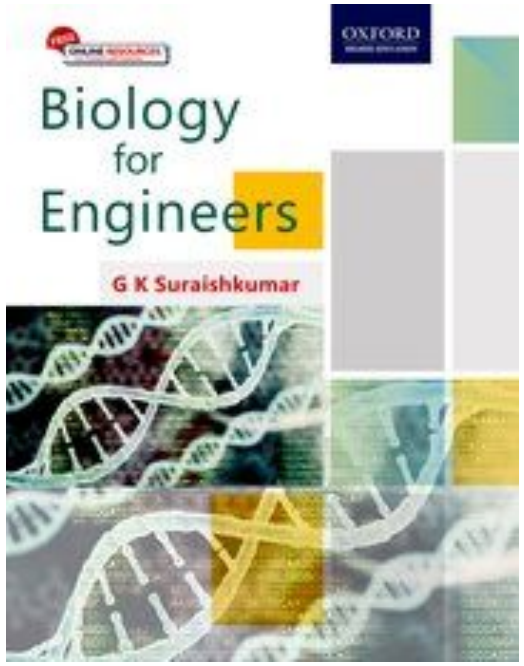
physics
astronomy
chemistry
earth sciences

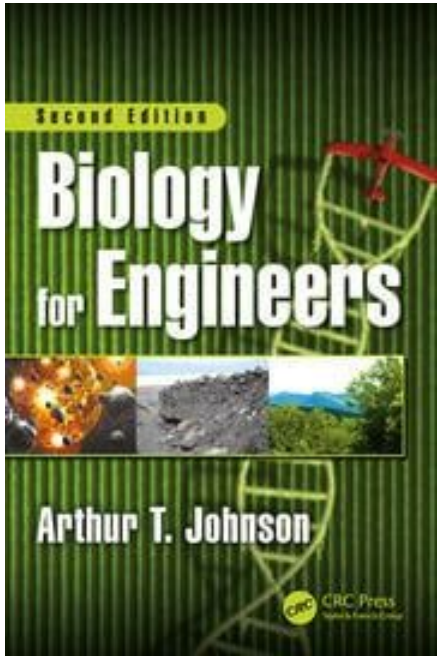
- Hence, having fundamental understanding of life sciences/Biology is important for better job prospects!

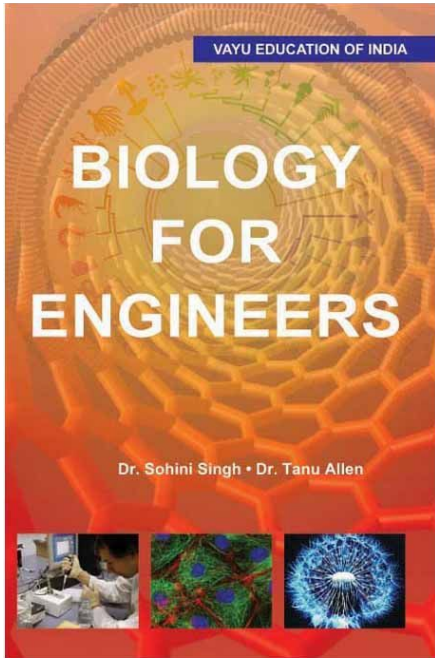
- So many books on “Biology for Engineers”





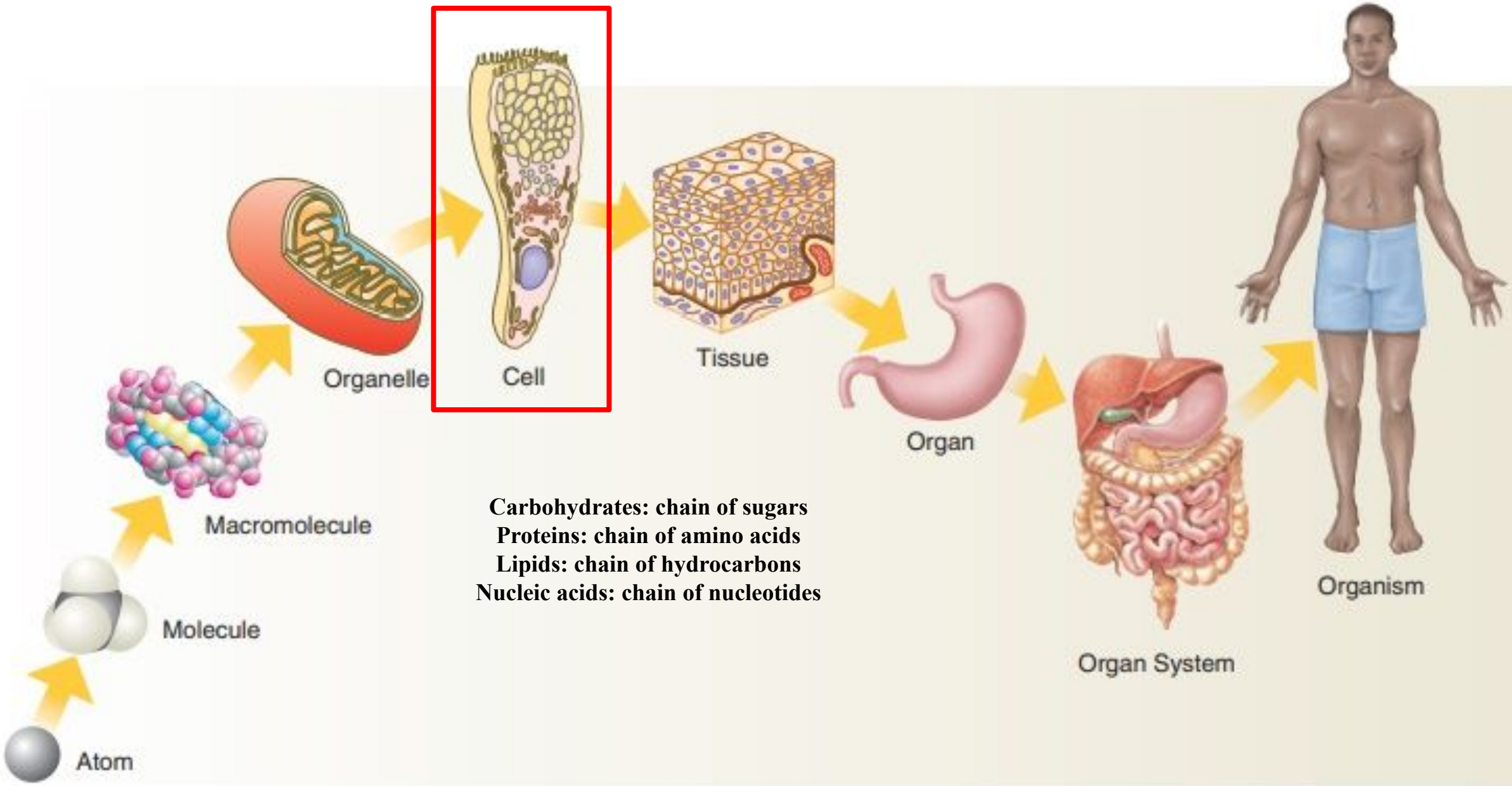






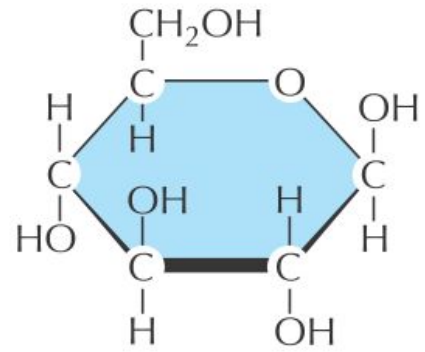
Organization level of the body

Basic unit of life

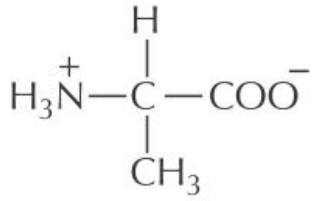


Carbohydrates: chain of sugars
Proteins: chain of amino acids
Lipids: chain of hydrocarbons
Nucleic acids: chain of nucleotides

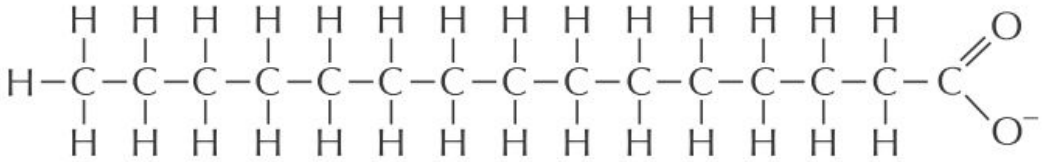
Building blocks of the macromolecules



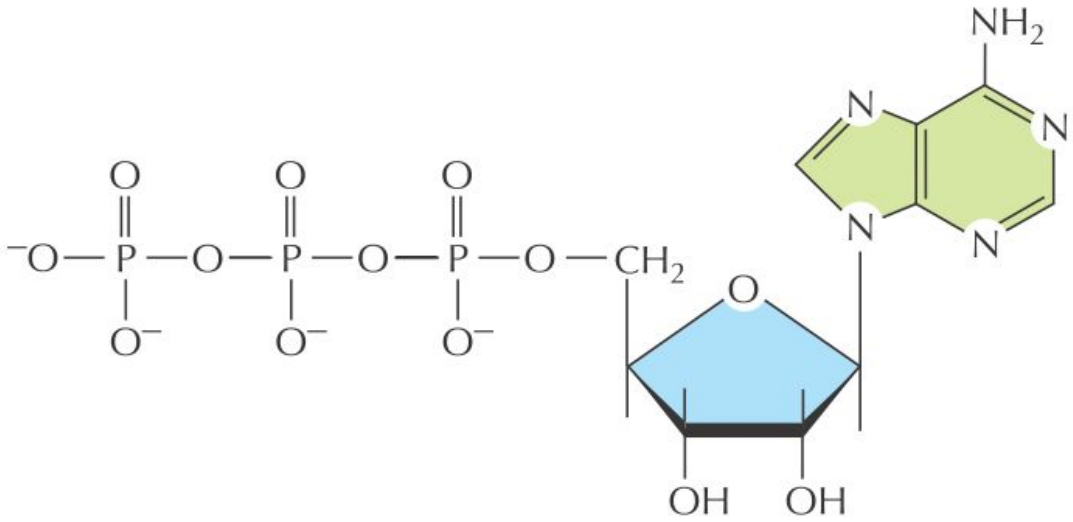
A SUGAR



AN AMINO ACID



A FATTY ACID



A NUCLEOTIDE

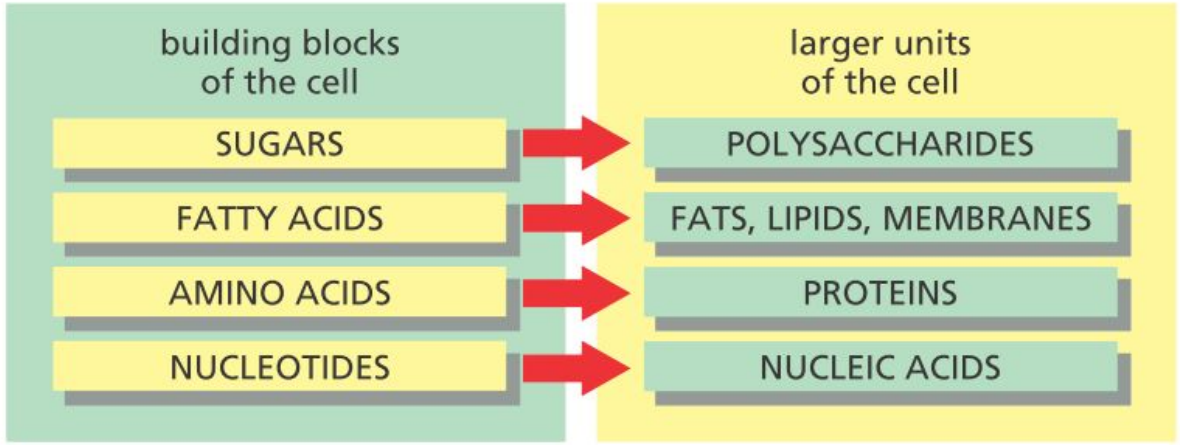
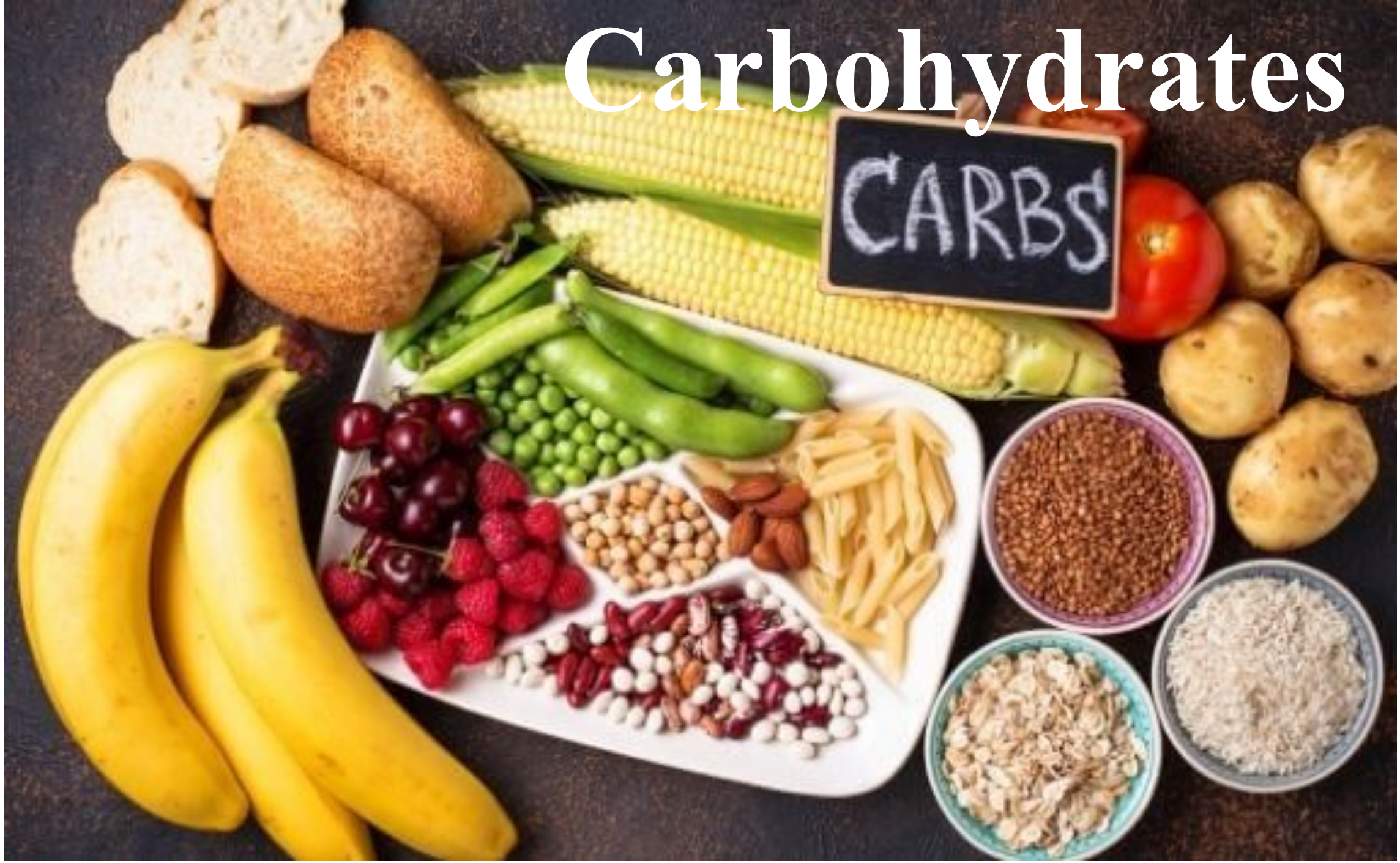


Figure 2–6 The four main families of small organic molecules in cells. These small molecules form the monomeric building blocks, or subunits, for most of the macromolecules and other assemblies of the cell. Some, such as the sugars and the fatty acids, are also energy sources. Their structures are outlined here and shown in more detail in the Panels at the end of this chapter and in Chapter 3.

Carbohydrates



What are carbohydrates?



NUTRITIONAL INFORMATION
(TYPICAL VALUES PER 100ml)

Energy	65kcal	Added Sugar	13.3g
Protein	0g	Fat	0g
Carbohydrate	16.2g	Vitamin A	120mcg
Natural Fruit Sugars	2.5g	Calcium	8mg

16.2 gm x 4 = 65 cal



* Nutritional Information per 100g	
1. Energy	562 kcal
2. Carbohydrate	46.10 g
Sugar	0.25 g
3. Total Fat	38.12 g
Saturated Fatty Acids	16.50 g
Monounsaturated Fatty Acids	13.64 g
Polyunsaturated Fatty Acids	3.10 g
Trans Fatty Acids	0 g
4. Protein	8.68 g
5. Fiber	0.83 g
6. Sodium	615 mg
7. Cholesterol	0 mg

* These are approximate values

46.10 gm x 4 = 184.4 cal
8.68 gm x 4 = 34.72 cal
38.12 gm x 9 = 343.08 cal
Total = 562 cal

1 gm of carbohydrates provide 4 calories.
1 gm of proteins provide 4 calories
1 gm of fats/lipids provide 9 calories

Note: Nutritional calorie is the equivalent of the kilocalorie used by chemists and biochemists. So, 65 calorie diet is the equivalent of a 65 kcal diet.

What are carbohydrates?

- Carbohydrates, or carbs, are sugar molecules. Along with proteins and fats, carbohydrates are one of three main nutrients found in foods and drinks.

- Carbohydrates are an essential macronutrient the body requires in large amounts to run smoothly.

- Our digestive system breaks down carbohydrates into glucose. Glucose, or blood sugar, is the main source of energy for your body's cells, tissues, and organs. Glucose can be used immediately or stored in the liver and muscles for later use.

Which foods have carbohydrates?

- Grains, such as bread, noodles, pasta, crackers, cereals, and rice
- Fruits, such as apples, bananas, berries, mangoes, melons, and oranges
- Dairy products, such as milk and yogurt
- Legumes, including dried beans, lentils, and peas
- Snack foods and sweets, such as cakes, cookies, candy, and other desserts
- Juices, regular sodas, fruit drinks, sports drinks, and energy drinks that contain sugar
- Starchy vegetables, such as potatoes, corn, and peas

Some foods don't have a lot of carbohydrates, such as meat, fish, poultry, some types of cheese, nuts, and oils.

Classification of Carbohydrates

Carbohydrates are classified by structure and named from their composition: **carbon**, **hydrogen**, and **oxygen**.

SIMPLE CARBOHYDRATES

Monosaccharides

SINGLE SUGAR UNITS

Primary Examples:

- Glucose
- Fructose
- Galactose

Classification by Functional Group:

Aldose Sugars

Contain aldehyde group (-CHO)

• Glucose, Galactose

Keto Sugars

Contain ketone group (C=O)

• Fructose

Disaccharides

TWO JOINED MONOSACCHARIDES

Maltose

Malt Sugar

Glucose + Glucose

Lactose

Milk Sugar

Glucose + Galactose

Sucrose

Table Sugar

Glucose + Fructose

COMPLEX CARBOHYDRATES

Polysaccharides

LONG CHAINS OF MONOSACCHARIDES

Starches

Energy storage form in plants (e.g., grains, potatoes, legumes).

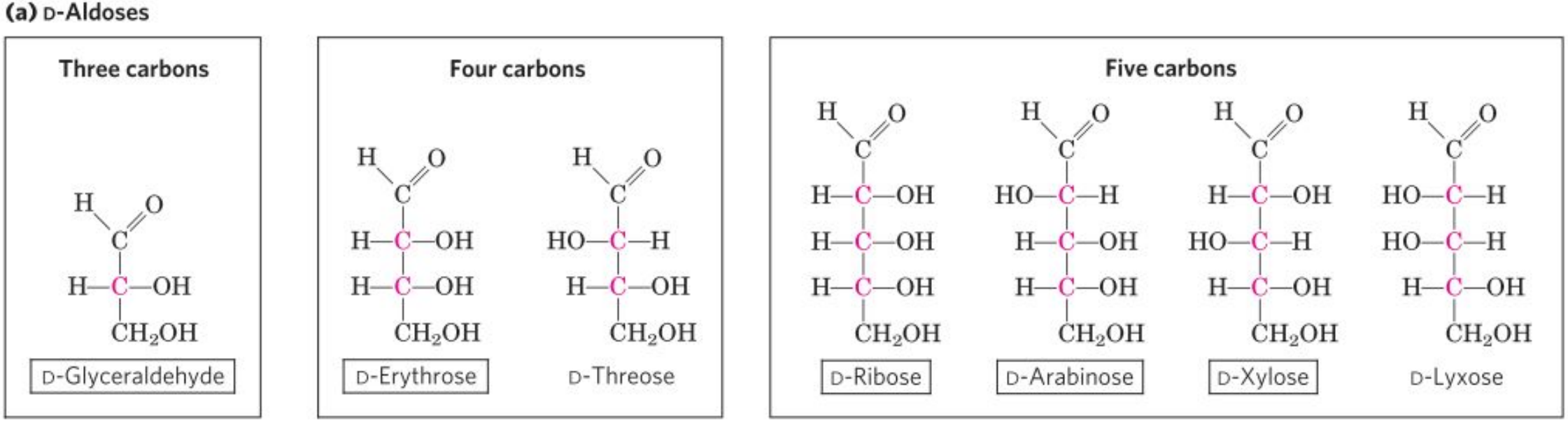
Fibers

Structural plant material; non-digestible carbohydrate providing roughage.

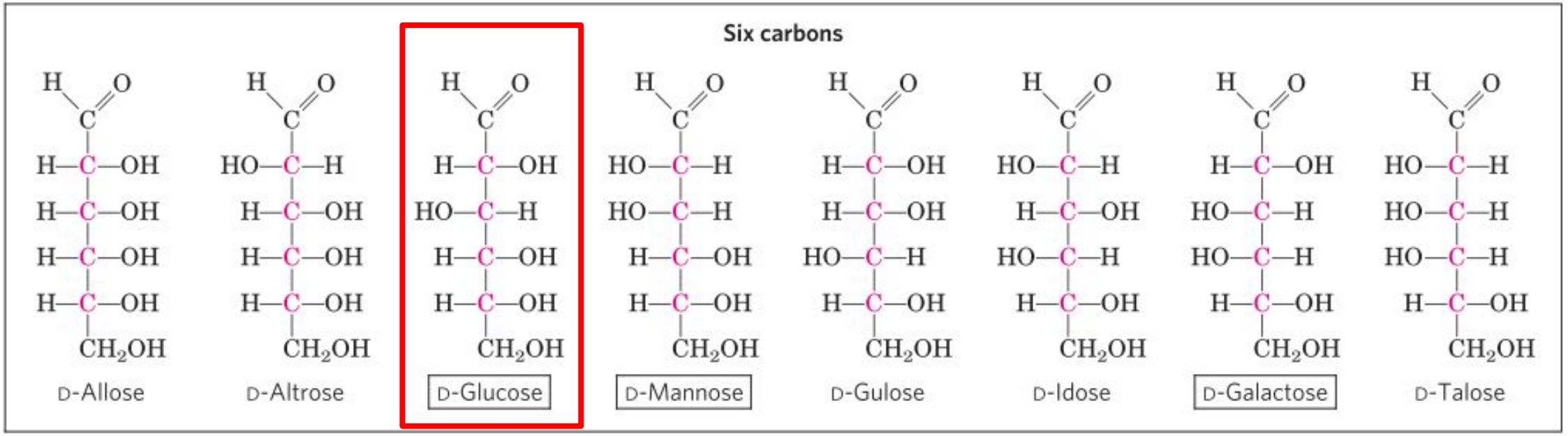
Glycogen

Primary energy storage glucose polymer stored in liver and muscle tissue.

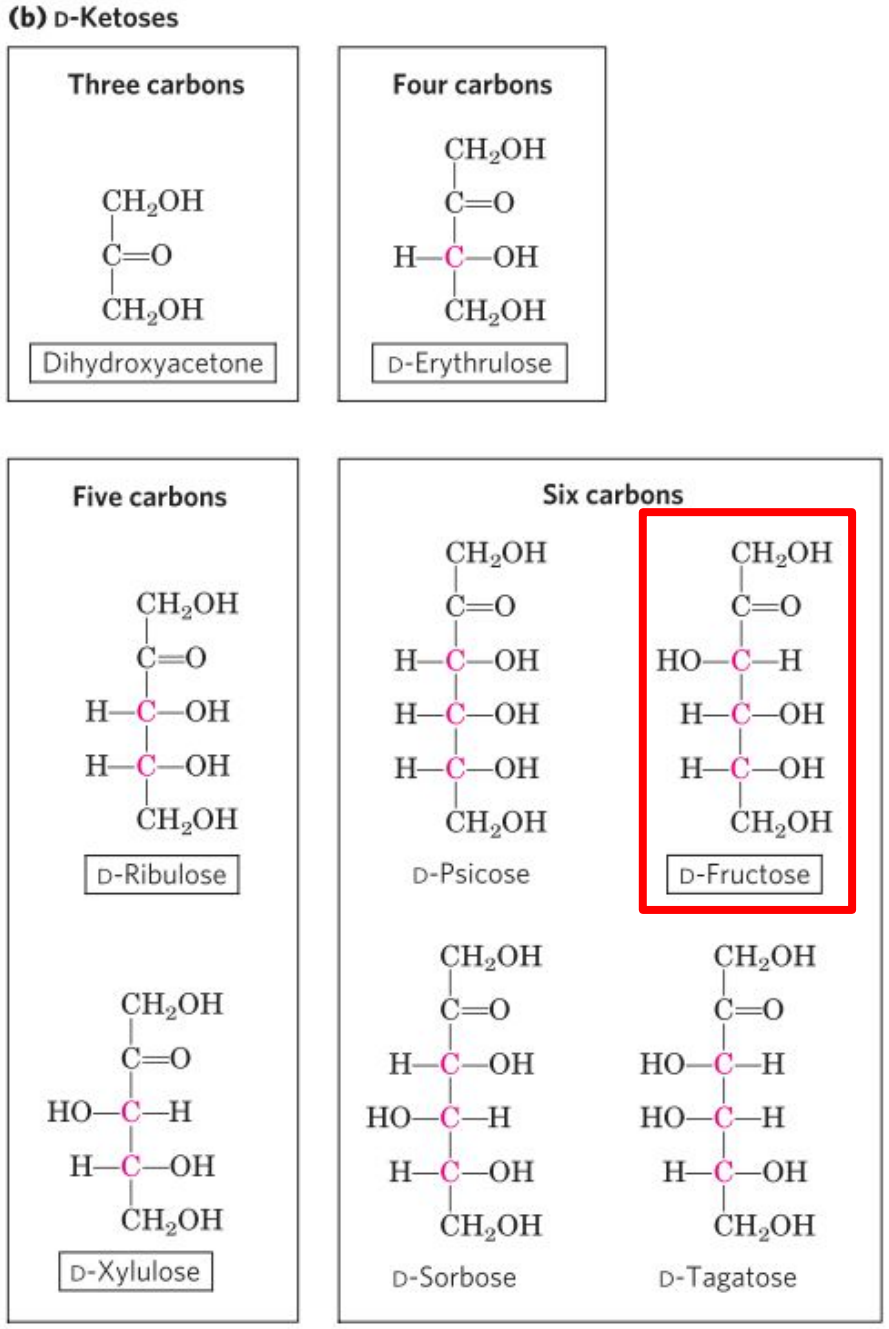
Structures of carbohydrates_Aldose sugars



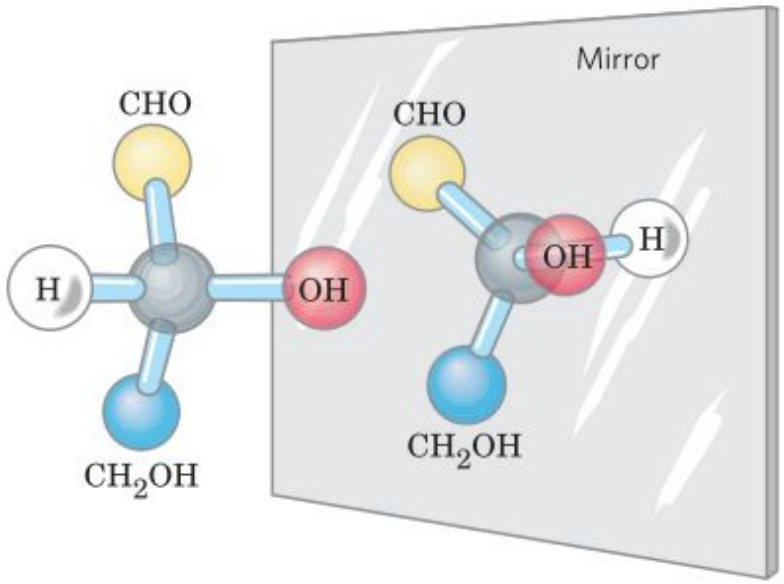
Chiral carbon atoms are shown in pink.



Structures of carbohydrates_Keto sugars



Stereoisomers of monosaccharides



Ball-and-stick models



Fischer projection formulas



Enantiomers are the mirror images of each other.

Enantiomers of Glyceraldehyde

A molecule with n chiral centers can have 2^n stereoisomers.

Glyceraldehyde has 2^1 =2,
Aldohexoses with 4 chiral carbons have 2^4=16.

FIGURE 7–2 Three ways to represent the two enantiomers of glyceraldehyde. The enantiomers are mirror images of each other. Ball-and-stick models show the actual configuration of molecules. Recall (see Fig. 1–18) that in perspective formulas, the wide end of a solid wedge projects out of the plane of the paper, toward the reader; a dashed wedge extends behind.

Epimers of glucose

Monosaccharides generally contain several chiral carbons and exist in a variety of stereochemical forms.

Glucose
$C_6H_{12}O_6$

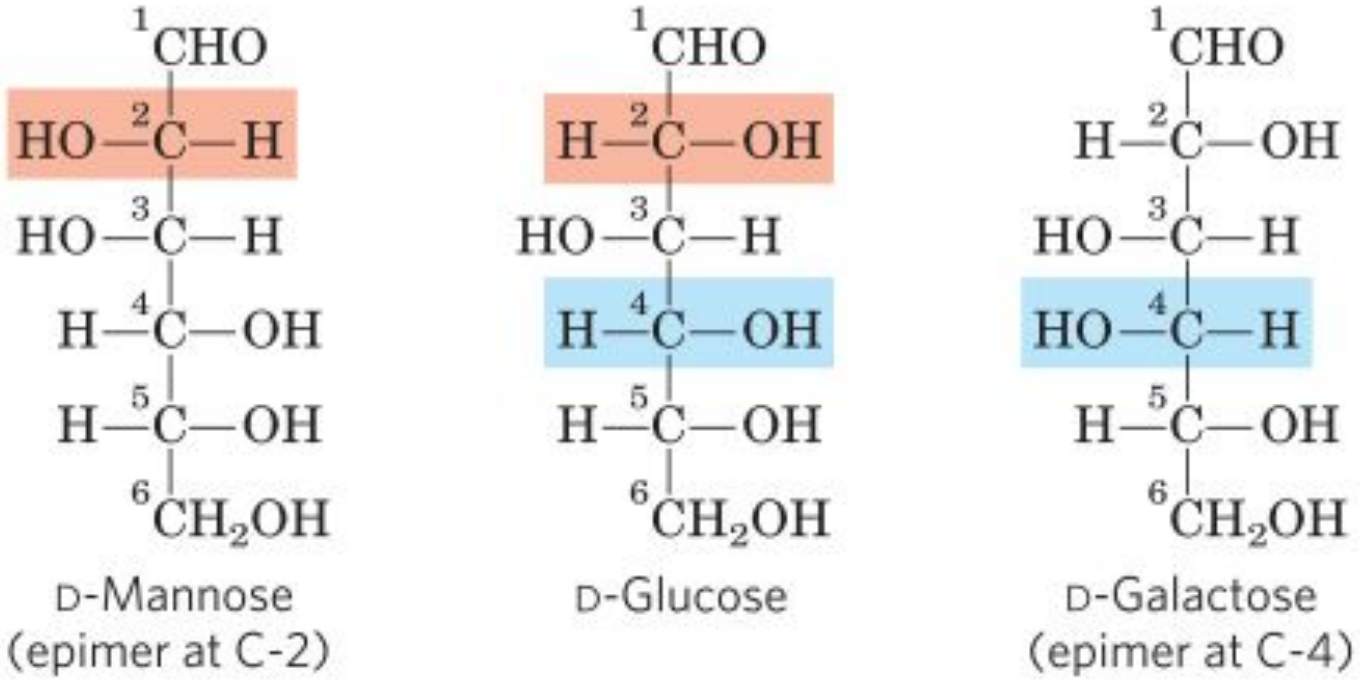
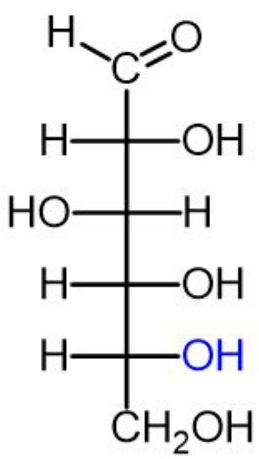


FIGURE 7–4 Epimers. D-Glucose and two of its epimers are shown as projection formulas. Each epimer differs from D-glucose in the configuration at one chiral center (shaded light red or blue).

Chiral carbon atoms are the carbon atoms that are attached to four substituents/groups.

Enantiomer and Epimer of D-glucose

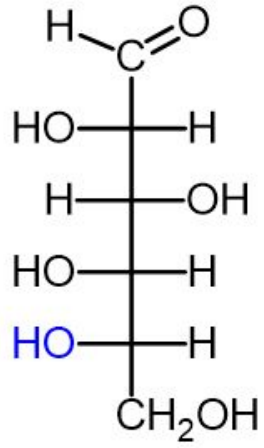


D-glucose

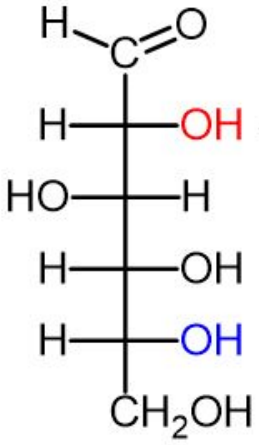


Enantiomers

All the chiral centers have different configuration



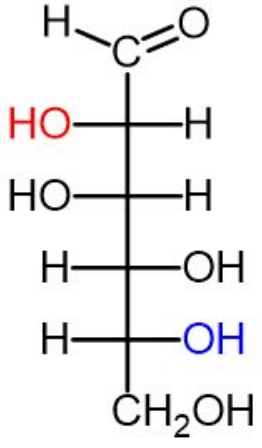
L-glucose



D-Glucose

Only one chiral center with different configuration

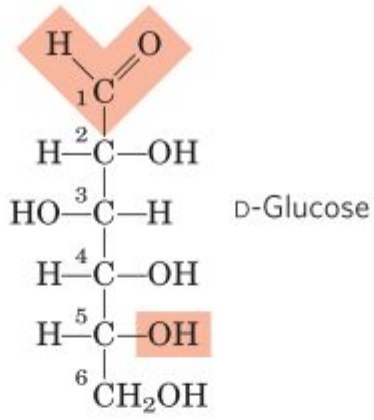
Epimers

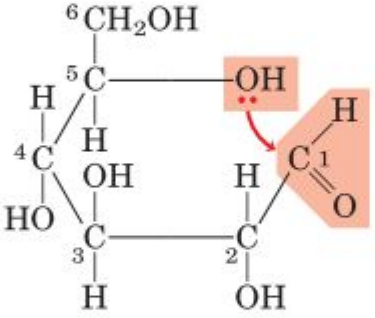


D-Mannose

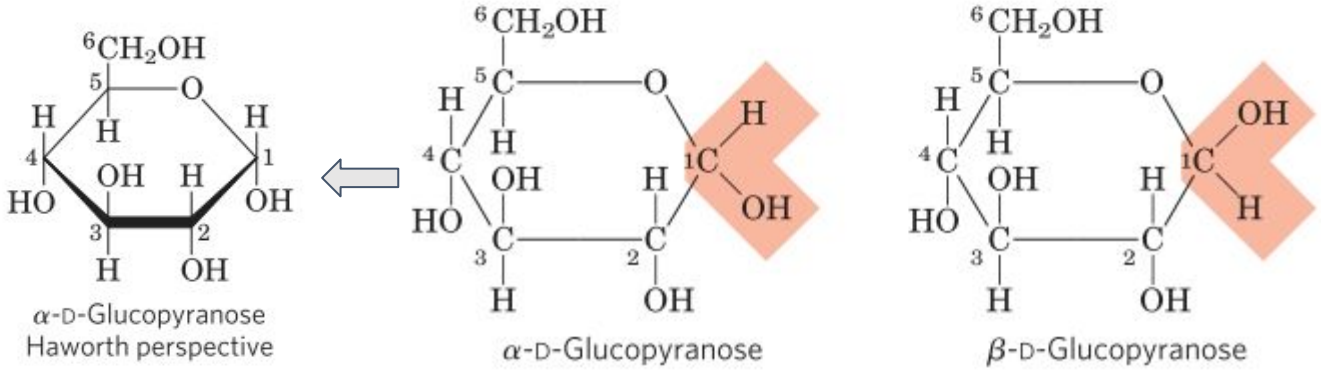
Cyclic (ring) structures of monosaccharides

- In aqueous solution, monosaccharides with 5 or more carbons occur predominantly as cyclic (ring) structures.
- Carbonyl group forms a covalent bond with the oxygen of a hydroxyl group along the chain.
- Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called anomers, and the carbonyl carbon is called the anomeric carbon.
- Mutarotation: The interconversion of α- and β- anomers is called mutarotation. This is a spontaneous process.





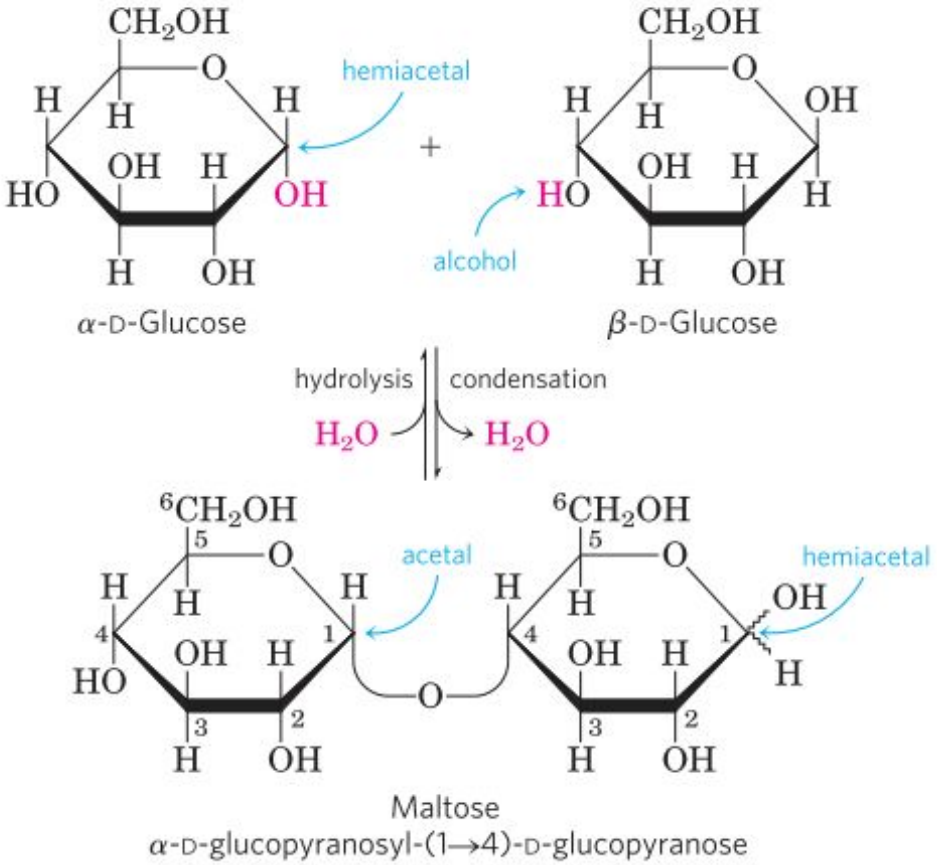
Anomers



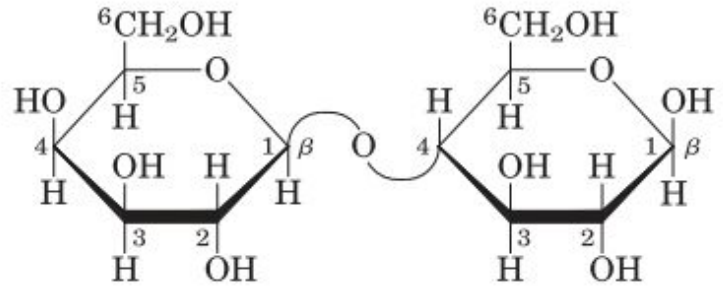
Disaccharides contain a glycosidic bond

- Disaccharides (such as maltose, lactose and sucrose) consist of two monosaccharides joined covalently by an O-glycosidic bond, which is formed when a hydroxyl group of one sugar reacts with the anomeric carbon of the other.

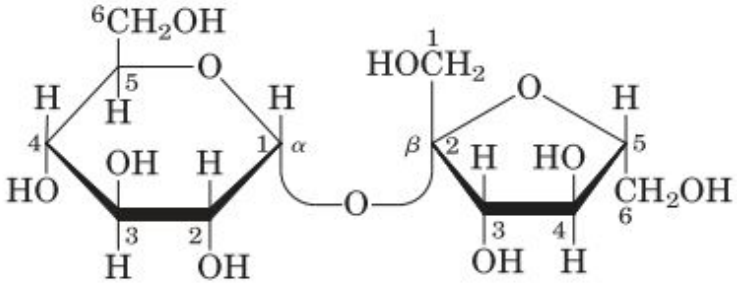
- Anomeric carbon is involved in a glycosidic bond, hence the easy interconversion of linear and cyclic forms is prevented.



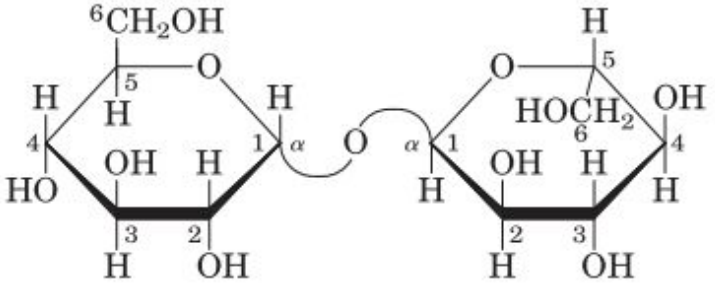
Disaccharides and their structures



Lactose (β form)
β-D-galactopyranosyl-(1→4)-β-D-glucopyranose
Gal(β1→4)Glc



Sucrose
β-D-fructofuranosyl α-D-glucopyranoside
Fru(2β↔α1)Glc ≡ Glc(α1↔2β)Fru

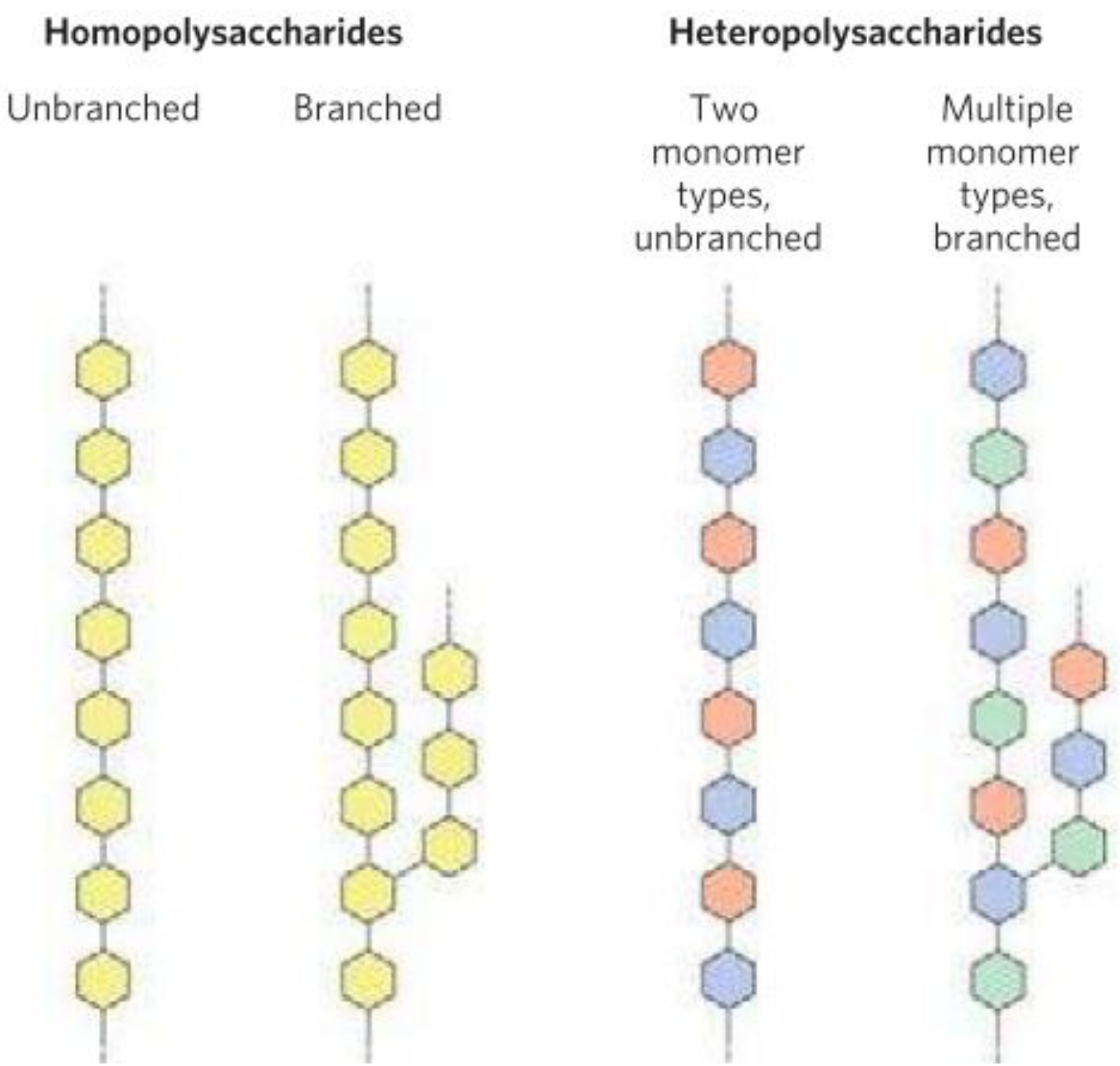


Trehalose
α-D-glucopyranosyl α-D-glucopyranoside
Glc(α1↔1α)Glc

Polysaccharides

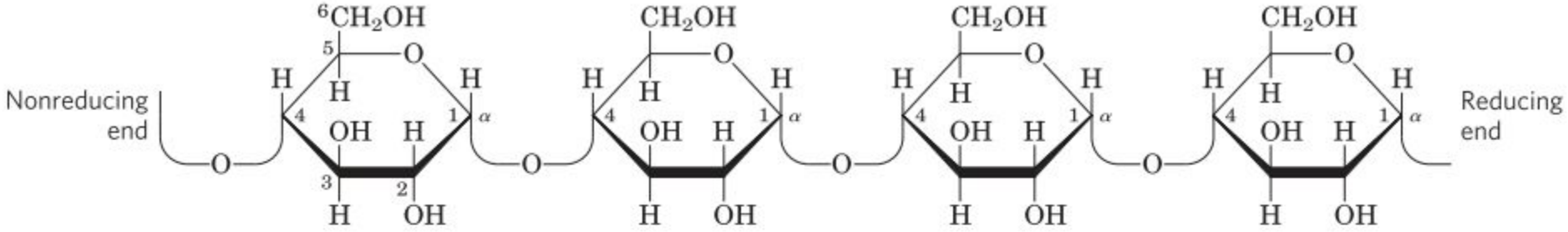
- Most carbohydrates found in nature occur as polysaccharides, polymer of medium to high molecular weight.
- Polysaccharides are also known as glycans.
- They serve as stored fuel (e.g. starch, glycogen) and as structural components of cell walls (cellulose, chitin, dextran) and extracellular matrix.
- Homopolysaccharides contain only a single monomeric species. e.g. starch, glycogen, cellulose, chitin, dextran.
- Heteropolysaccharides contain two or more different kinds. e.g. peptidoglycan,

Homo- and Hetero- polysaccharides



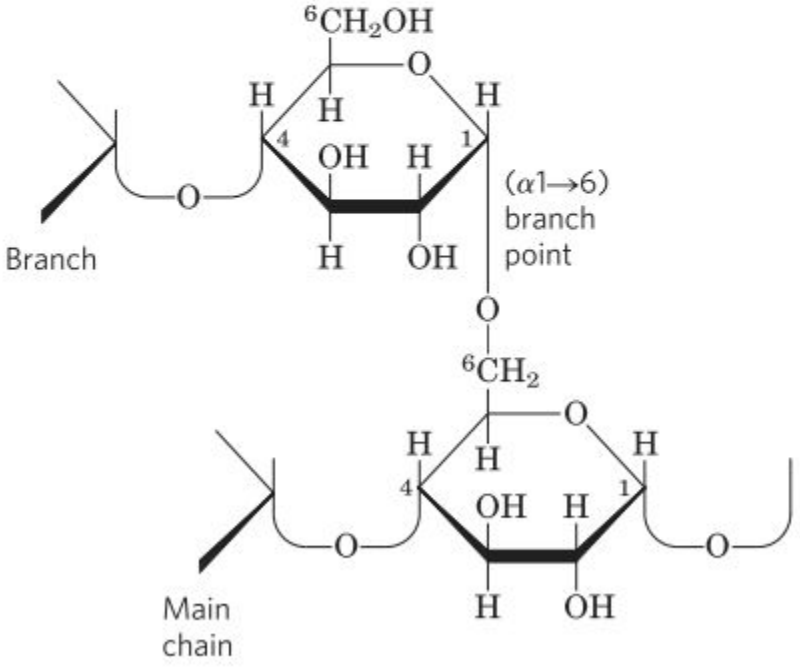
Homo- and Hetero- polysaccharides

A short segment of Amylose, a linear polymer of D-glucose residues in α1→4 linkage.



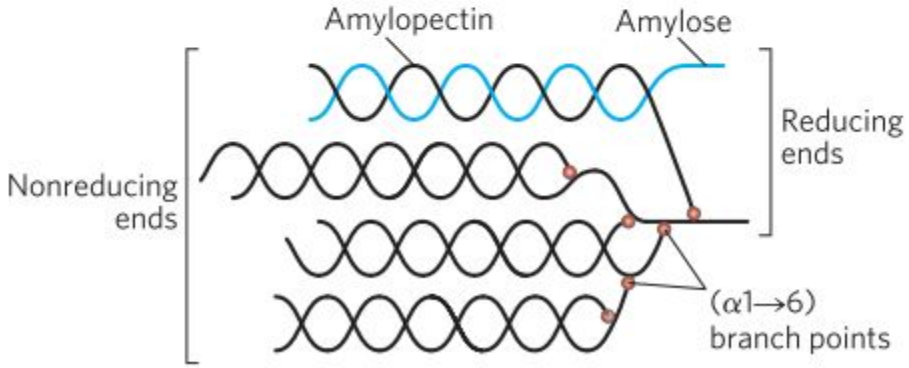
(a) Amylose

An α1→6 branch point of glycogen or amylopectin.



(b)

A cluster of amylose and amylopectin in starch granules.

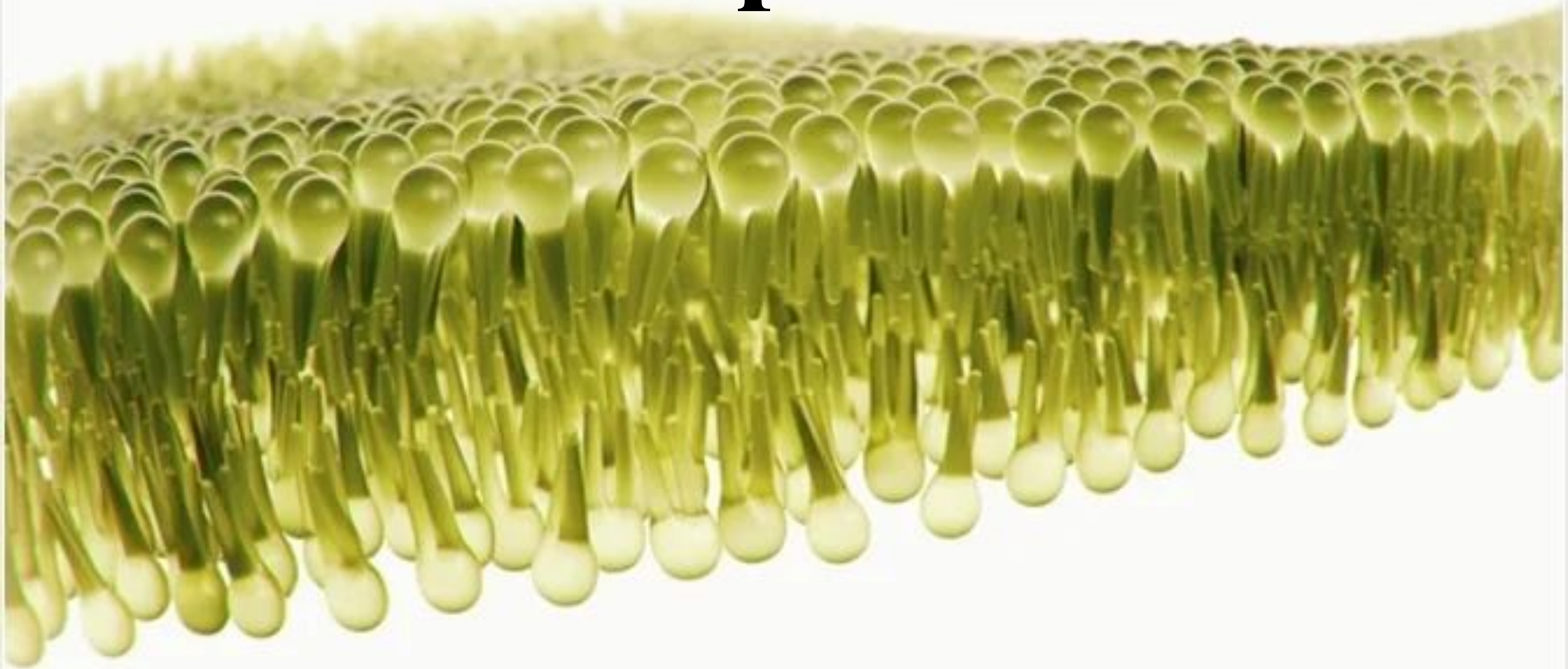


(c)

TABLE 7–2 Structures and Roles of Some Polysaccharides

Polymer	Type*	Repeating unit†	Size (number of monosaccharide units)	Roles/significance
Starch				Energy storage: in plants
Amylose	Homo-	(α1→4)Glc, linear	50–5,000	
Amylopectin	Homo-	(α1→4)Glc, with (α1→6)Glc branches every 24–30 residues	Up to 10^6	
Glycogen	Homo-	(α1→4)Glc, with (α1→6)Glc branches every 8–12 residues	Up to 50,000	Energy storage: in bacteria and animal cells
Cellulose	Homo-	(β1→4)Glc	Up to 15,000	Structural: in plants, gives rigidity and strength to cell walls
Chitin	Homo-	(β1→4)GlcNAc	Very large	Structural: in insects, spiders, crustaceans, gives rigidity and strength to exoskeletons
Dextran	Homo-	(α1→6)Glc, with (α1→3) branches	Wide range	Structural: in bacteria, extracellular adhesive
Peptidoglycan	Hetero-; peptides attached	4)Mur2Ac(β1→4)GlcNAc(β1	Very large	Structural: in bacteria, gives rigidity and strength to cell envelope
Agarose	Hetero-	3)D-Gal(β1→4)3,6-anhydro-L-Gal(α1	1,000	Structural: in algae, cell wall material
Hyaluronan (a glycosamino-glycan)	Hetero-; acidic	4)GlcA(β1→3)GlcNAc(β1	Up to 100,000	Structural: in vertebrates, extracellular matrix of skin and connective tissue; viscosity and lubrication in joints

Lipids



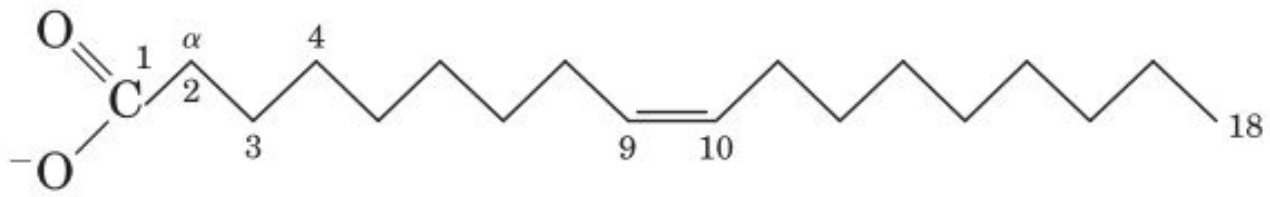
What are Lipids?

- Biological lipids are chemically diverse group of compounds, which are insoluble in water.
- Fats and oils are the stored form of energy.
- Phospholipids and sterols are major structural elements of biological membranes.
- Other lipids are hormones, enzyme cofactors, light absorbing pigments, electron carriers, intracellular messengers.

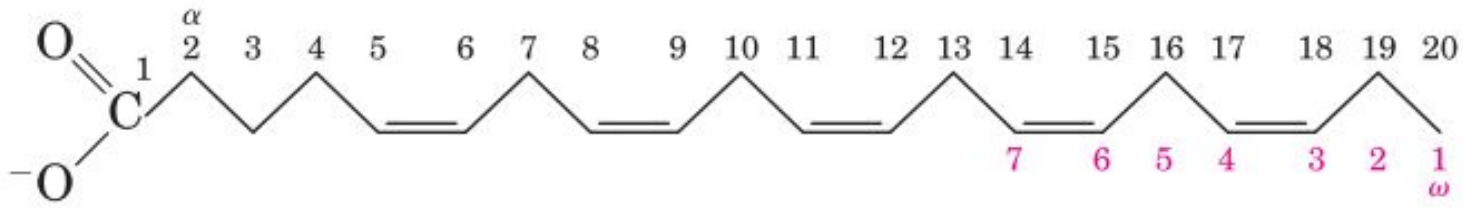
Fatty Acids

- The fats and oils used as stored forms of energy are derivatives of fatty acids.
- Fatty acids are hydrocarbon derivatives. They are carboxylic acids with hydrocarbon chains ranging from 4 to 36 carbons long (C4 to C36).
- In some fatty acids, this chain is unbranched and fully saturated (no double bonds).

16-carbon saturated palmitic acid is abbreviated 16:0
18-carbon oleic acid, with one double bond is abbreviated 18:1.



(a) 18:1(Δ^9) *cis*-9-Octadecenoic acid



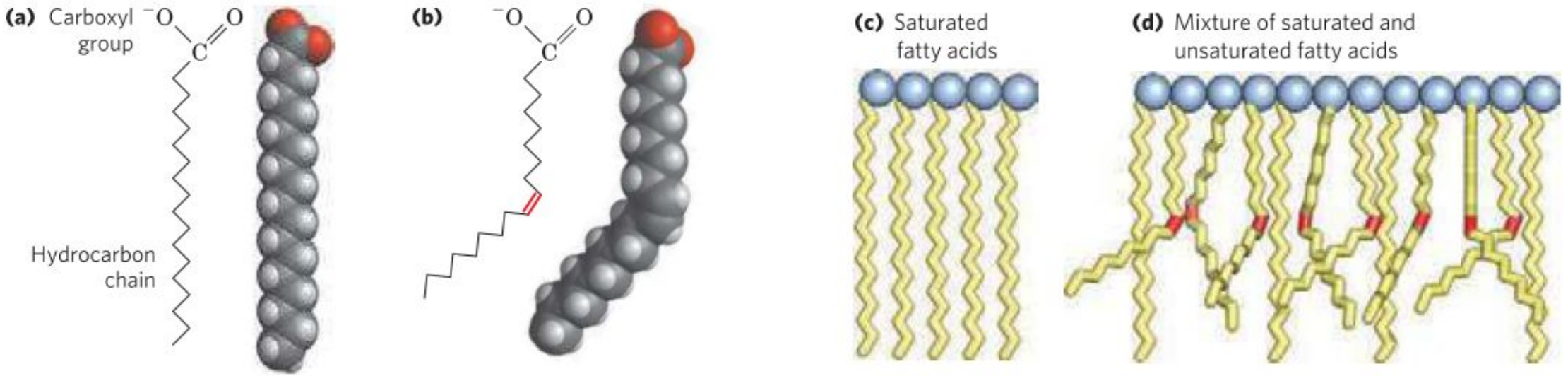
(b) 20:5($\Delta^{5,8,11,14,17}$) Eicosapentaenoic acid (EPA), an omega-3 fatty acid

PUFA: Polyunsaturated fatty acid

KEY CONVENTION: The family of **polyunsaturated fatty acids (PUFAs)** with a double bond between the third and fourth carbon from the methyl end of the chain are of special importance in human nutrition. Because the physiological role of PUFAs is related more to the position of the first double bond near the *methyl* end of the chain than to the carboxyl end, an alternative nomenclature is sometimes used for these fatty acids. The carbon of the methyl group—that is, the carbon most distant from the carboxyl group—is called the ω (omega) carbon and is given the number 1 (Fig. 10–1b). In this convention, PUFAs with a double bond between C-3 and C-4 are called **omega-3 (ω-3) fatty acids**, and those with a double bond between C-6 and C-7 are **omega-6 (ω-6) fatty acids**. ■

Physical properties of the fatty acids

- The physical properties of the fatty acids are determined by the length and degree of unsaturation of the hydrocarbon chain.
- The nonpolar hydrocarbon chain accounts for the poor solubility of fatty acids in water.
- **The longer the fatty acyl chain and the fewer the double bonds, the lower is the solubility in water.**
- The carboxylic acid group is polar and accounts for the slight solubility of short-chain fatty acids in water.
- Melting points are also influenced by the length and degree of unsaturation of hydrocarbon chain.
- At room temperature, the saturated fatty acids from 12:0 to 24:0 have a waxy consistency, whereas unsaturated fatty acids are oily liquids.
- In unsaturated fatty acids, the double bond forces a kink in the hydrocarbon chain, hence they cannot pack together as tightly as fully saturated fatty acids.



Common types of storage and membrane lipids

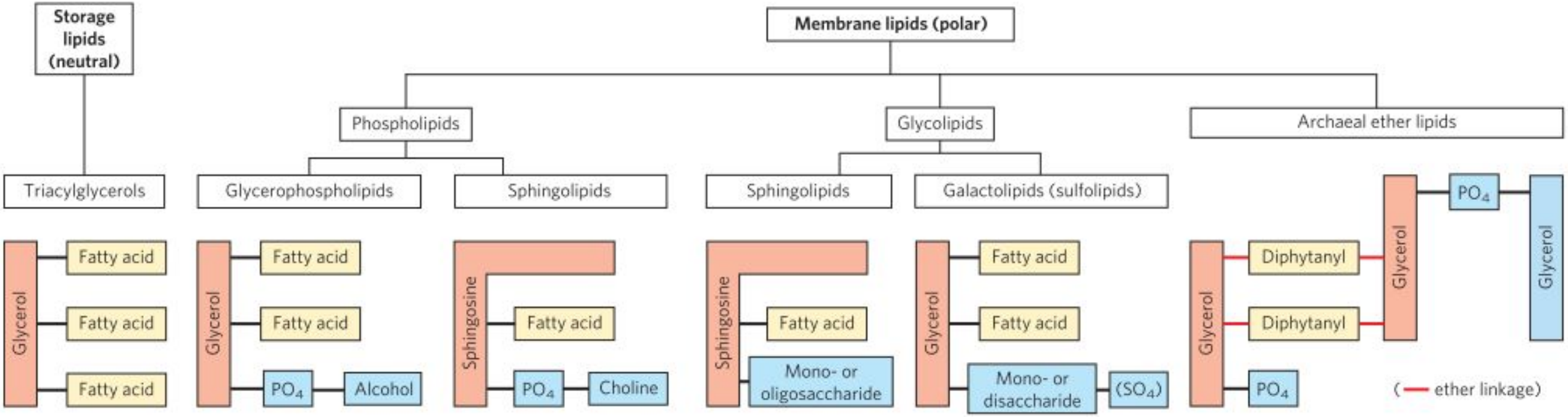
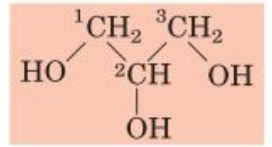


TABLE 10–3 Eight Major Categories of Biological Lipids

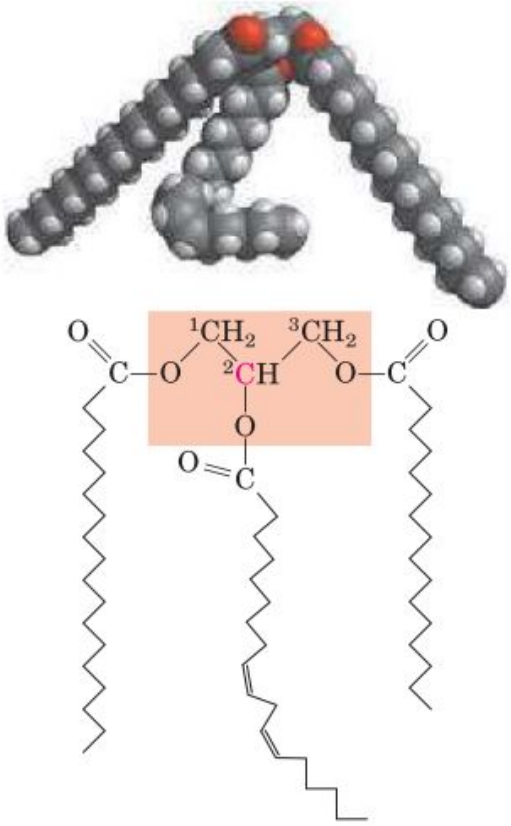
Category	Category code	Examples
Fatty acids	FA	Oleate, stearoyl-CoA, palmitoylcarnitine
Glycerolipids	GL	Di- and triacylglycerols
Glycerophospholipids	GP	Phosphatidylcholine, phosphatidylserine, phosphatidylethanolamine
Sphingolipids	SP	Sphingomyelin, ganglioside GM2
Sterol lipids	ST	Cholesterol, progesterone, bile acids
Prenol lipids	PR	Farnesol, geraniol, retinol, ubiquinone
Saccharolipids	SL	Lipopolysaccharide
Polyketides	PK	Tetracycline, erythromycin, aflatoxin B_1

Triacylglycerols are fatty acid esters of glycerol and they serve as stored energy

- The simplest lipids constructed from fatty acids are the triacylglycerols, also referred to as triglycerides, fats or neutral fats.
- Triacylglycerols are composed of three fatty acids each in ester linkage with a single glycerol.
- Those containing same fatty acids in all three positions are called simple triacylglycerols.
  16:0, 18:0 and 18:1 are tripalmitin, tristearin and triolein.
- Triacylglycerols provide stored energy and they are stored in adipocytes (fat cells). They are also stored as oils in the seed of many plants, provides energy during seed germination.
- Humans have fat tissue (adipocytes) under the skin, in the abdominal cavity and in the mammary glands. Moderately obese people with 15-20 kg of triacylglycerols deposited in their adipocytes could meet their energy needs for months. In contrast, the human body can store less than a day's energy supply in the form of glycogen.



Glycerol



1-Stearoyl, 2-linoleoyl, 3-palmitoyl glycerol, a mixed triacylglycerol

Advantages of storing energy as triacylglycerols compared to polysaccharides

1) Oxidation of triacylglycerols provides more than twice the energy compared to glycogen:
   1 gm of carbohydrates provides 4 calories.
   1 gm of triacylglycerols provides 9 calories.

2) As triacylglycerols are hydrophobic (unhydrated), the organism that carries fat as fuel doesn't have to carry the extra weight of water of hydration that is associated with stored polysaccharides (2 g per gram of polysaccharide).

TABLE 10–1 Some Naturally Occurring Fatty Acids: Structure, Properties, and Nomenclature

Carbon skeleton	Structure*	Systematic name†	Common name (derivation)	Melting point (°C)	Solubility at 30 °C (mg/g solvent)	
					Water	Benzene
12:0	$CH_3(CH_2)_{10}COOH$	*n*-Dodecanoic acid	Lauric acid (Latin *laurus*, "laurel plant")	44.2	0.063	2,600
14:0	$CH_3(CH_2)_{12}COOH$	*n*-Tetradecanoic acid	Myristic acid (Latin *Myristica*, nutmeg genus)	53.9	0.024	874
16:0	$CH_3(CH_2)_{14}COOH$	*n*-Hexadecanoic acid	Palmitic acid (Latin *palma*, "palm tree")	63.1	0.0083	348
18:0	$CH_3(CH_2)_{16}COOH$	*n*-Octadecanoic acid	Stearic acid (Greek *stear*, "hard fat")	69.6	0.0034	124
20:0	$CH_3(CH_2)_{18}COOH$	*n*-Eicosanoic acid	Arachidic acid (Latin *Arachis*, legume genus)	76.5		
24:0	$CH_3(CH_2)_{22}COOH$	*n*-Tetracosanoic acid	Lignoceric acid (Latin *lignum*, "wood" + *cera*, "wax")	86.0		
16:1(Δ^9)	$CH_3(CH_2)_5CH{=}CH(CH_2)_7COOH$	*cis*-9-Hexadecenoic acid	Palmitoleic acid	1 to –0.5		
18:1(Δ^9)	$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$	*cis*-9-Octadecenoic acid	Oleic acid (Latin *oleum*, "oil")	13.4		
18:2($\Delta^{9,12}$)	$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	*cis*-,*cis*-9,12-Octadecadienoic acid	Linoleic acid (Greek *linon*, "flax")	1–5		
18:3($\Delta^{9,12,15}$)	$CH_3CH_2CH{=}CHCH_2CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	*cis*-,*cis*-,*cis*-9,12,15-Octadecatrienoic acid	α-Linolenic acid	–11		
20:4($\Delta^{5,8,11,14}$)	$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CHCH_2CH{=}CHCH_2CH{=}CH(CH_2)_3COOH$	*cis*-,*cis*-,*cis*-,*cis*-5,8,11,14-Icosatetraenoic acid	Arachidonic acid	–49.5		

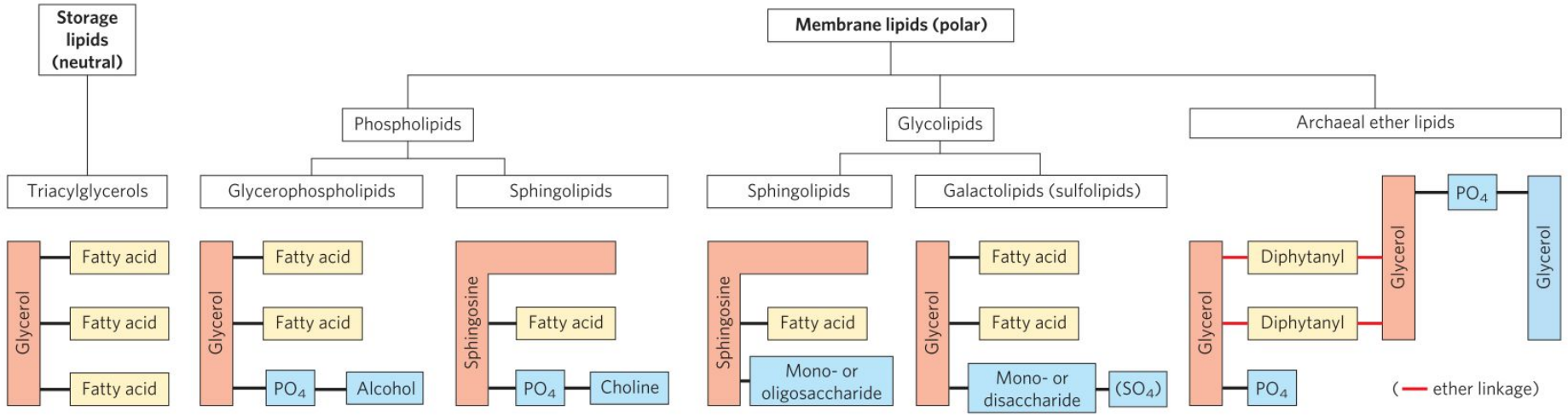
Structural Lipids in Membranes

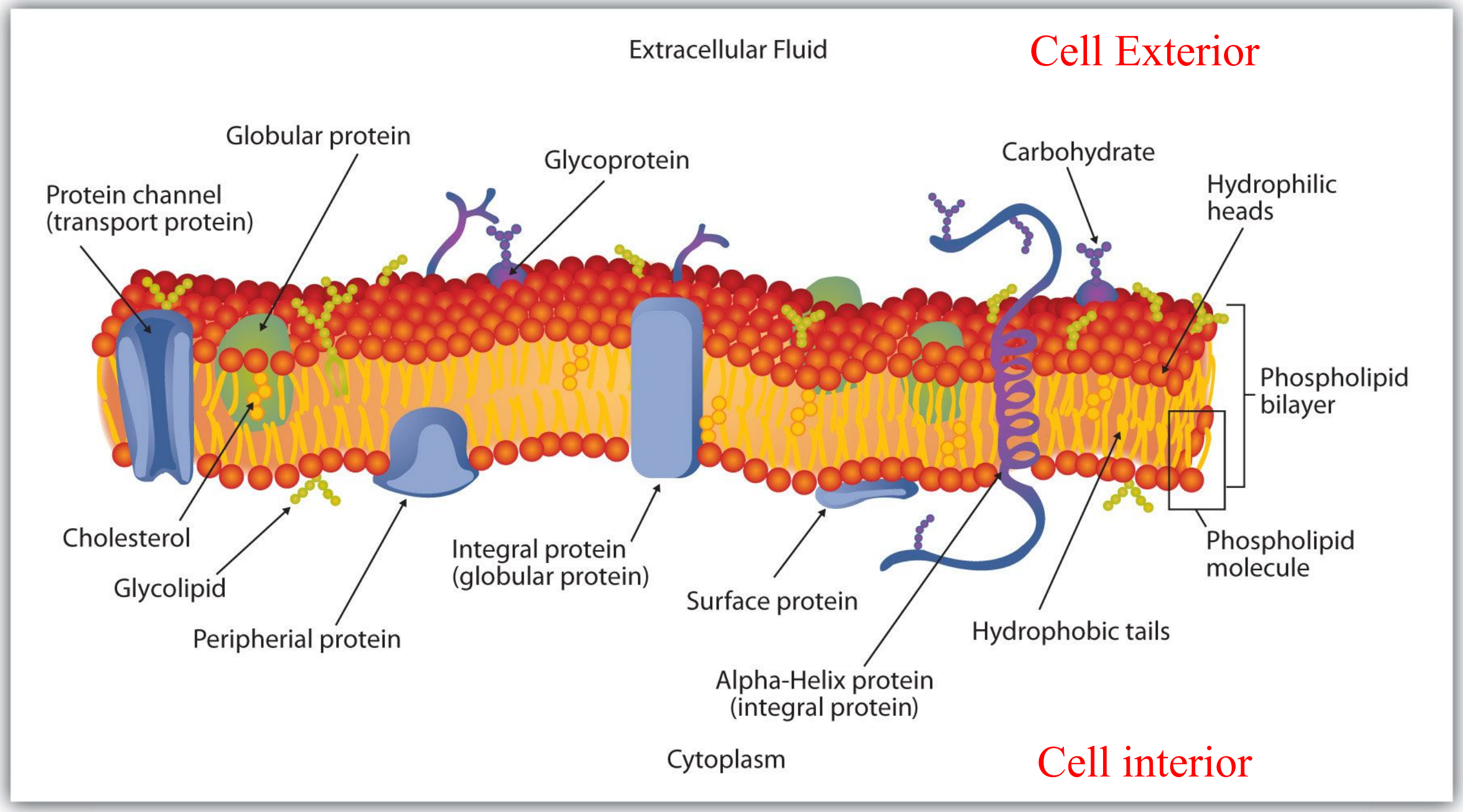
Biological membranes (cell membrane, nuclear membrane) are made up of lipid bilayers.

Membrane lipids are amphipathic: one end is hydrophobic and the other end is hydrophilic.

Five types of membrane lipids:

1) glycerophospholipids: two fatty acids joined to glycerol.
2) galactolipids and sulfolipids: two fatty acids esterified to glycerol, but lack the phosphate of phospholipids.
3) tetraether lipids: two very long alkyl chains are ether-linked to glycerols at both ends
4) sphingolipids: a single fatty acid is joined to fatty amine, sphingosine
5) sterols: rigid system of four fused hydrocarbon rings.




Extracellular Fluid
Cell Exterior
Globular protein
Protein channel
(transport protein)
Glycoprotein
Carbohydrate
Hydrophilic
heads
Phospholipid
bilayer
Cholesterol
Glycolipid
Peripherial protein
Integral protein
(globular protein)
Surface protein
Phospholipid
molecule
Hydrophobic tails
Alpha-Helix protein
(integral protein)
Cytoplasm
Cell interior

Why omega-3 PUFA are required in your diet?

Humans require but do not have the enzymatic capacity to synthesize the omega-3 PUFA α-linolenic acid (ALA; 18:3($\Delta^{9,12,15}$), in the standard convention), and must therefore obtain it in the diet. From ALA, humans can synthesize two other omega-3 PUFAs important in cellular function: eicosapentaenoic acid (EPA; 20:5($\Delta^{5,8,11,14,17}$), shown in Fig. 10–1b) and docosahexaenoic acid (DHA; 22:6($\Delta^{4,7,10,13,16,19}$)). An imbalance of omega-6 and omega-3 PUFAs in the diet is associated with an increased risk of cardiovascular disease. The optimal dietary ratio of omega-6 to omega-3 PUFAs is between 1:1 and 4:1, but the ratio in the diets of most North Americans is closer to 10:1 to 30:1. The "Mediterranean diet," which has been associated with lowered cardiovascular risk, is richer in omega-3 PUFAs, obtained in leafy vegetables (salads) and fish oils. The latter oils are especially rich in EPA and DHA, and fish oil supplements are often prescribed for individuals with a history of cardiovascular disease. ■

Partial Hydrogenation of cooking oils produces trans fatty acids



- When oily foods are exposed too long to the oxygen in air, they may spoil and become rancid.
- The unpleasant smell and taste associated with rancidity result from the oxidative cleavage of dc unsaturated fatty acids, which produces aldehydes and carboxylic acids of shorter chain length.
- To improve the shelflife of vegetable oils used in cooking, partial hydrogenation is used which c double bonds of fatty acids to single bonds.
- Partial hydrogenation has another undesirable effect: some cis double bonds are converted to trans double bonds.
- Dietary intake of trans fats leads to higher incidences of cardiovascular disease.
- Dietary trans fats raise the level of triacylglycerols and of LDL (“bad”) cholesterol in the blood and lower the level of HDL (“good”) cholesterol.
- Many fast foods are deep-fried in partially hydrogenated vegetable oils and therefore contain high levels of trans fatty acids. Several countries have restricted the use of partially hydrogenated oils in restaurants.

TABLE 10–2 Trans Fatty Acids in Some Typical Fast Foods and Snacks

	Trans fatty acid content	
	In a typical serving (g)	As % of total fatty acids
French fries	4.7–6.1	28–36
Breaded fish burger	5.6	28
Breaded chicken nuggets	5.0	25
Pizza	1.1	9
Corn tortilla chips	1.6	22
Doughnut	2.7	25
Muffin	0.7	14
Chocolate bar	0.2	2

Glycosphingolipids as determinants of blood groups

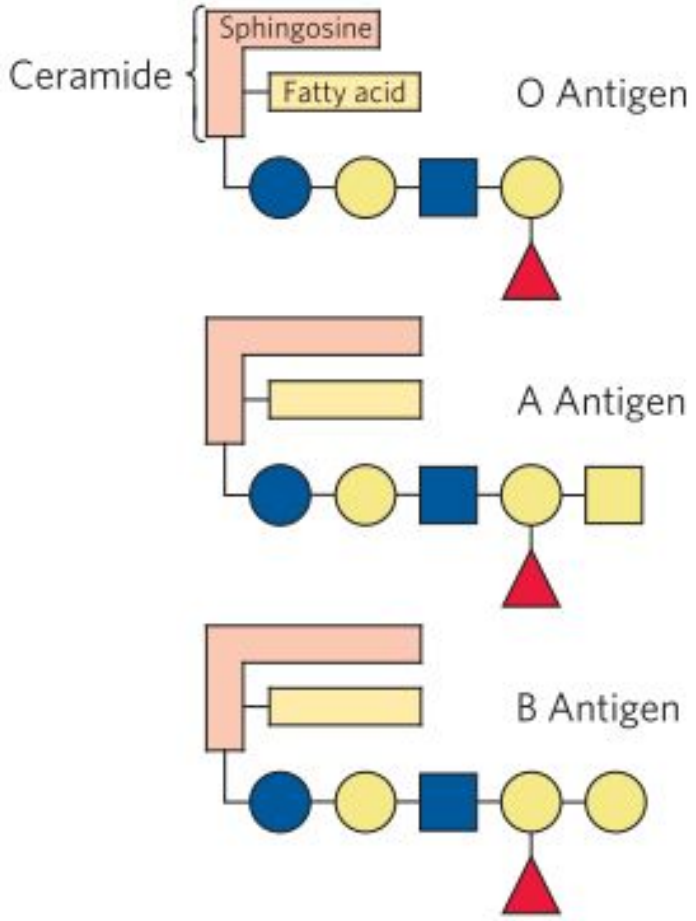


FIGURE 10–15 Glycosphingolipids as determinants of blood groups. The human blood groups (O, A, B) are determined in part by the oligosaccharide head groups of these glycosphingolipids. The same three oligosaccharides are also found attached to certain blood proteins of individuals of blood types O, A, and B, respectively. Standard symbols for sugars are used here (see Table 7–1).

Glycosphingolipids are present on the outer surface of the red blood cells (RBCs)

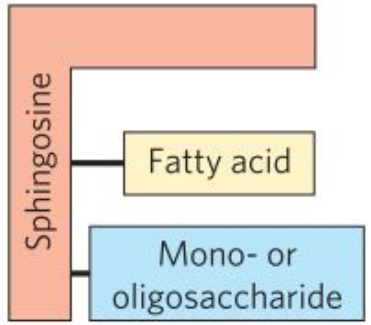


TABLE 7–1 Symbols and Abbreviations for Common Monosaccharides and Some of Their Derivatives

Abequose	Abe	Glucuronic acid	GlcA
Arabinose	Ara	Galactosamine	GalN
Fructose	Fru	Glucosamine	GlcN
Fucose	Fuc	*N*-Acetylgalactosamine	GalNAc
Galactose	Gal	*N*-Acetylglucosamine	GlcNAc
Glucose	Glc	Iduronic acid	IdoA
Mannose	Man	Muramic acid	Mur
Rhamnose	Rha	*N*-Acetylmuramic acid	Mur2Ac
Ribose	Rib	*N*-Acetylneuraminic acid (a sialic acid)	Neu5Ac
Xylose	Xyl		

Further reading

Book: Lehninger: Principles of Biochemistry, 6th edition. ISBN: 1464109621.

Chapter 7: Carbohydrates and Glycobiology, Page 243-280.

Chapter 10: Lipids, Page 357-383.

Which types of carbohydrates you should eat?

You do need to eat some carbohydrates to give your body energy. But it's important to eat the right kinds of carbohydrates for your health:

- When eating grains, choose mostly **whole grains** and **not refined grains**:
  - Whole grains are foods like whole wheat bread, brown rice, whole cornmeal, and oatmeal. They offer lots of nutrients that your body needs, like vitamins, minerals, and fiber.
  - Refined grains are foods that have had some parts of the grains removed. This also removes some of the nutrients that are good for your health.
- Eat foods with **lots of fiber**. The Nutrition Facts labels on the back of food packages tells you how much fiber a product has.
- Try to avoid foods that have a lot of **added sugar**. These foods can have many calories but not much nutrition. **Eating too much added sugar raises your blood sugar and can make you gain weight.**

What are different types of carbohydrates?

- **Sugars:** simple carbohydrates. They can be added to foods, such as the sugar in candy, desserts, processed foods, and regular soda. They also include the kinds of sugar that are found naturally in fruits, vegetables, and milk.

- **Starches:** complex carbohydrates. Your body needs to break starches down into sugars to use them for energy. Starches include bread, cereal, and pasta. They also include certain vegetables, like potatoes, peas, and corn.

- **Fiber:** complex carbohydrate. Your body cannot break down most fibers, so eating foods with fiber can help you feel full and make you less likely to overeat. Diets high in fiber have other health benefits. They may help prevent stomach or intestinal problems, such as constipation. They may also help lower cholesterol and blood sugar. Fiber is found in many foods that come from plants, including fruits, vegetables, nuts, seeds, beans, and whole grains.